The Accuracy of the Paleomagnetic Method As Evaluated from Historic Hawaiian Lava Flows

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Abstract. An analysis was made of the various levels at which errors enter in paleomagnetic studies of laval flows, using as controls lava flows from Hawaii which cooled in known fields. The largest intraflow angular variations occur between different individually oriented cores and are due to rotation of blocks after cooling, to intraflow magnetic field variations at the time of cooling, and to orientation errors; other sources of angular error have much smaller effects. The ultimate accuracy of a paleomagnetic study is limited by the magnetic anomalies existing over the old terrain where the lava flows cooled. Because of these ancient anomalies, little additional accuracy can be gained in volcanic regions like Hawaii by collecting more than six oriented samples per flow, a number which results in an accuracy at the 95 per cent confidence level of 3° to 5°. If these results are applied to older lavas on Hawaii, the low rates of secular variation previously noted from observatory data are found to persist back to 1750.

INTRODUCTION

Nine lava flows extruded at known dates between 1750 and 1955 were sampled as part of a paleomagnetic study of 152 Quaternary lava flows from the island of Hawaii [*Doell and Cox*, 1961]. Since the regional geomagnetic field in Hawaii is known from direct observations beginning in 1900, the three youngest flows (1955, 1935, and 1907) are valuable as experimental controls to establish the accuracy with which directions of the regional geomagnetic field can be determined paleomagnetically. Within the limits of accuracy thus established, the results from the six older flows extend the record of geomagnetic secular variation on Hawaii from 1900, the date of the earliest direct observatory measurements, back to 1750, the age of the oldest historically dated lava flow on Hawaii.

EXPERIMENTAL PROCEDURE

Distribution of samples. The historic flows were sampled at sites distributed over the entire area of recent volcanism on Hawaii, the most widely separated flows being over 100 km apart (Figure 1). Five to ten oriented samples were collected at each sampling site over a lateral distance of from 6 to 1310 meters. Although most of these historic flows are exposed over much greater distances, sampling was restricted to facilitate comparison with the prehistoric flows.

Coring and orientation. Oriented samples were collected as cores 2.5 cm in diameter and from 10 to 20 cm long. Coring was done in situ with a small gasoline-powered, water-cooled diamond coring drill. To reduce the possibility of systematic errors, cores were drilled along many different directions at each sampling site. While still attached to the outcrop the cores were oriented by means of a nonmagnetic orienting tool which scribed a line on the upper side of the core

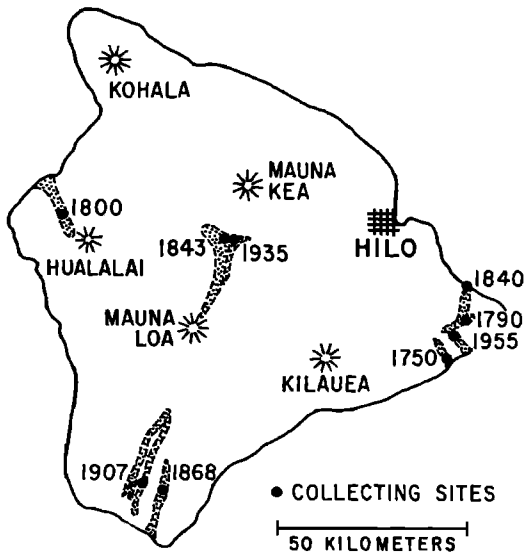


Fig. 1. Locations of historic lava flows on Hawaii sampled for this study.

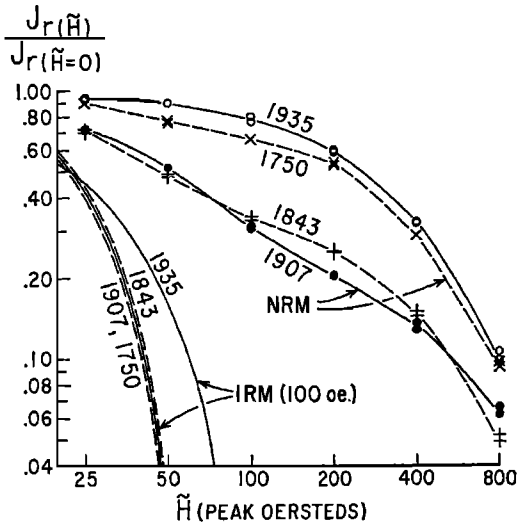


Fig. 2. Demagnetization curves of natural remanent magnetization (NRM) and isothermal remanent magnetization acquired in a 100-oe constant field (IRM). $J_r(\tilde{H})$ is the intensity remaining after treatment in an alternating magnetic field of $\tilde{H}$ oe peak value.

exactly parallel to the core axis. The plunge of the orientation line and the geographic azimuth of a horizontal line perpendicular to it, determined by triangulation, were recorded. The accuracy of this orientation technique is estimated from mechanical considerations to be ±2° (s.d.). Specimens 2.28 cm long were cut from the cores for measurement of remanent magnetization.

Measurement of remanent magnetization. Remanent magnetization was measured on an induction or 'spinner' magnetometer [*Johnson and McNish,* 1938; *Nagata,* 1961]. All data were reduced graphically using an equal-area projection accurate to 0.1° [*Doell and Altenhofen,* 1960] and were later checked analytically using a digital computer. Measurements of direction of magnetization have a precision of ±0.7° (s.d.) and an estimated accuracy of ±1.3° (s.d.) (see below). Intensity measurements have a precision of ±1 per cent (s.d.) and an estimated accuracy of ±6 per cent (s.d.).

Demagnetization in alternating fields. The apparatus used for demagnetization experiments consisted of a Helmholtz coil 11.5 cm in diameter supplied with current from the 60-cps mains. The Helmholtz coil was connected in series with capacitors to form a 60-cps resonant circuit, and in parallel with a 120-cps pass filter to suppress even harmonics [*As and Zijderveld,* 1958]. Alternating current in the coil was controlled by a variable transformer which lowered the current nearly linearly without step decreases from the peak value to zero in 10 minutes. As the current was decreased, the specimen was rotated simultaneously about three mutually perpendicular axes in an apparatus made of nonmagnetic and nonconducting material [*Creer,* 1959]. The inner rotation axis was parallel to the axis of the specimen cylinder and was rotated at 386 rpm; through a set of gears this axis in turn was rotated at 225 rpm about a second axis perpendicular to the first. The second axis was rotated at 160 rpm about a third axis parallel to the axis of the coil. The results of the experiments were unchanged either when the rates of rotation of the axes were changed, keeping the gear ratios fixed, or when the ambient magnetic field was nulled [*Graham,* 1961]. The apparatus was designed to minimize undesirable magnetizations that may be added by the demagnetization process. However, as a check on its efficiency, all demagnetization experiments were repeated at each value of the peak demagnetizing field, the specimen being rotated 180° in the demagnetization apparatus between runs.

Results

Stability of natural remanent magnetization. Specimens from four different flows that occurred between 1750 and 1935 were demagnetized in alternating magnetic fields with peak intensities of 25, 50, 100, 200, 400, and 800 oe. The average value of the fraction of the original

TABLE 1. Intensities of NRM and IRM of Specimens Used for Demagnetization Curves Shown in Figure 2 (Units: 10^{-3} emu/cm³)

Date of Flow	J_{NRM}	$J_{NRM}(\tilde{H}$ = 100 oe)	$J_{IRM}(h$ = 100 oe)	$J_{IRM}(h$ = 2200 oe)
1935	1.98	1.56	8.35	90
1907	6.11	1.90	86.0	390
1843	6.71	2.25	81.0	301
1750	8.16	5.40	68.0	442

$\tilde{H}$ = peak value of alternating demagnetization field.
h = strength of constant field inducing IRM.

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natural remanent magnetization (NRM) remaining after two successive demagnetizations at each of these intensities is shown in Figure 2. From 30 to 80 per cent of the NRM remains after demagnetization in 100-oe alternating fields (Table 1), and 5 to 10 per cent remains after demagnetization at 800 oe, indicating that the NRM of these samples is extremely stable.

Figure 2 also shows demagnetization curves, obtained as before, for isothermal remanent magnetization (IRM) induced by a 100-oe constant field applied for about 2 seconds to these same specimens. The intensity of this IRM is from 5 to 14 times larger than the original NRM (Table 1), and the slopes of the IRM demagnetization curves are considerably steeper than those of the corresponding NRM curves. The experiments of *Rimbert* [1959] and *Cox* [1961] show that IRM produced by any constant field smaller than 100 oe would have a lower intensity and a steeper demagnetization curve. Therefore, since the demagnetization curves of the original NRM have no steep segments above 25 oe, it is safe to conclude that no component of IRM such as might have been induced by exposure to constant fields during drilling or shipping remains after demagnetization in alternating fields above 25 oersteds.

Components of magnetization other than thermoremanent magnetization (TRM), which was acquired during the initial cooling of these flows, are demonstrably absent, with the possible exception of a small amount of viscous magnetization. Geological evidence precludes reheating by subsequent volcanic activity. Chemical magnetization is negligible because petrological examination shows no indication of recrystallization of ferromagnetic minerals after initial cooling. IRM due to lightning is common in older rocks [*Cox*, 1961; *Graham*, 1961], but on Hawaii lightning is rare and almost certainly has not struck near these young flows, as the demagnetization experiments confirm. Although a component of viscous magnetization may be a part of all natural remanent magnetization, little time has been available for its development in these flows. The demagnetization curves shown in Figure 2 may serve, therefore, as useful standards for TRM formed under natural conditions.

A wide range of magnetic 'hardness' is shown by these curves. After demagnetization in a 200-oe alternating field, for example, the fraction of the NRM remaining varies by as much as a factor of 3 from flow to flow. Since this variation shows no correlation with the ages of the flows, it is not due to viscous magnetization. However, there is a correlation between magnetic hardness and rate of cooling. The specimens from the 1750 and 1935 eruptions were collected near the tops of flows, whereas the magnetically less hard specimens from the 1843 and 1907 eruptions were collected from the massive central parts of flows. Petrological data that also indicate a difference in cooling rates are listed in Table 2.

TABLE 2. Petrographic Description of Specimens Used for Demagnetization Curves Shown in Figure 2

Date of Flow	Texture	Groundmass	Opaques
1935	Nonporphyritic, abundant vesicles (3.2–0.1 mm).	Fine to very fine microlites (0.22–0.05 mm), glass common.	Magnetite(?), very delicate skeletal crystals 0.006–0.045 mm (5.5%). Ilmenite (?), needles 0.012–0.03 mm (1.5%).
1750	Rare phenocrysts, abundant vesicles (10–0.15 mm).	Holocrystalline, very fine microlites (0.07–0.02 mm).	Magnetite(?), euhedral to irregular crystals 0.045–0.015 mm (7%). Ilmenite (?),needles 0.045–0.015 mm (3%).
1907	Abundant phenocrysts (2.0–0.15 mm), few vesicles (1.5–0.3 mm).	Holocrystalline intergranular texture, fine to very fine microlites (0.1–0.02 mm).	Magnetite(?), euhedral to irregular crystals 0.06–0.003 mm (10%).
1843	Abundant phenocrysts (1.3–0.15 mm), abundant vesicles (10–0.3 mm).	Fine-grained microlites (0.3–0.1 mm), minor glass in interstices.	Magnetite(?), euhedral, 0.006 mm to skeletal, 0.025-mm crystals (6%). Ilmenite(?), needles 0.045mm (2%).

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The hard magnetization of specimens from the tops of flows may be associated with higher coercive force domains because rapid cooling commonly results in smaller grain sizes which are known to have higher coercive forces [*Gottschalk*, 1935]. Moreover, crystal lattices may undergo dislocations during rapid cooling, and in multidomain magnetic crystals these highly strained regions may acquire TRM much as if they were single domains with very high coercive forces [*Verhoogen*, 1959]. Similar differences in demagnetization curves have been observed within a single submarine basalt layer [*Cox and Doell*, 1962]. Fine-grained specimens from near the quickly cooled top of this layer have curves similar to those of the hard specimens from the 1935 and 1750 Hawaiian flows, whereas the curves for coarser grained specimens from deeper in the same layer coincide rather closely with the less hard specimens from the 1907 and 1843 flows shown in Figure 2.

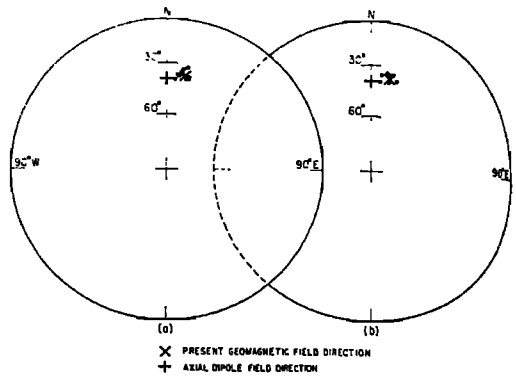


Fig. 3. Average direction of remanent magnetization of each flow before (*a*) and after (*b*) partial demagnetization in 100-oe alternating fields.

Directions of remanent magnetization. The average direction of the NRM of each flow is listed in Table 3 and plotted in Figure 3(*a*). Each flow average is based on measurements of

TABLE 3. Remanent Magnetization of Historic Flows

Date of Flow	N	Sample Spacing, meters	$\langle \Delta D \rangle \pm$ (s.d.)	$J \times 10^3$ ($\pm$ s.d.)	Decl. of J	Incl. of J	R	k_C	δ_C'	Δ	α_{95}	γ_{95}
1955	10	1310	0.6°W ± 3.1°	8.49 ± 2.21	7.1°	37.3°	9.17549	11	23.3°	3.2°	15.3°	15.4°
				6.45 ± 2.94	*11.5°*	*34.4°*	*9.95752*	*212*	*5.3°*	*2.4°*	*3.3°*	*3.9°*
1935	5	25	0.6°W ± 0.9°	5.61 ± 5.73	13.0°	37.5°	4.98265	230	4.8°	2.0°	5.0°	5.4°
				4.53 ± 4.60	*15.3°*	*37.5°*	*4.99164*	*479*	*3.3°*	*3.8°*	*3.5°*	*4.1°*
1907	6	56	0.0° ± 1.6°	7.68 ± 2.55	9.5°	35.9°	5.98408	314	4.2°	0.2°	3.8°	4.3°
				4.06 ± 2.06	*10.6°*	*36.2°*	*5.99302*	*716*	*2.8°*	*0.7°*	*2.5°*	*3.3°*
1868	6	34	0.0° ± 2.5°	8.04 ± 1.64	9.0°	33.4°	5.98917	462	3.4°		3.1°	3.7°
				6.12 ± 1.18	*9.1°*	*33.8°*	*5.97848*	*233*	*4.8°*		*4.4°*	*4.9°*
1843	7	35	0.2°W ± 2.5°	4.18 ± 1.56	9.6°	32.3°	6.97858	280	4.5°		3.6°	4.2°
				2.18 ± 0.73	*10.6°*	*35.0°*	*6.96374*	*165*	*5.8°*		*4.7°*	*5.1°*
1840	10	70	0.4°E ± 2.1°	4.63 ± 2.95	6.7°	36.6°	9.93235	133	6.7°		4.2°	4.7°
				3.49 ± 1.93	*6.1°*	*35.6°*	*9.92547*	*121*	*7.0°*		*4.4°*	*4.9°*
1800	7	44	5.4°E ± 1.8°	10.14 ± 2.19	11.3°	36.6°	6.94347	106	7.3°		5.9°	6.3°
				5.98 ± 3.47	*12.2°*	*34.8°*	*6.98821*	*509*	*3.3°*		*2.7°*	*3.4°*
1790	9	793	1.1°E ± 2.6°	7.55 ± 1.37	12.0°	38.9°	8.98592	568	3.2°		2.2°	3.0°
				6.16 ± 1.18	*11.0°*	*39.0°*	*8.99013*	*811*	*2.7°*		*1.8°*	*2.7°*
1750	7	6	0.9°E ± 4.2°	9.56 ± 2.16	5.2°	39.5°	6.97074	205	5.2°		4.2°	4.7°
				6.29 ± 2.66	*5.4°*	*39.7°*	*6.96635*	*178*	*5.6°*		*4.5*	*5.0°*

Notes: Italics indicate values after partial demagnetization at 100 oersteds.
N = number of sample cores.
$\langle \Delta D \rangle$ = average local declination error at sample sites.
J = intensity of remanent magnetization.
R = vector sum of the N unit vectors.
k_C = $(N - 1)/(N - R)$: the intraflow precision parameter [*Fisher*, 1953].
δ'_C = intraflow angular standard deviation (see equation 2).
Δ = angle between J and regional geomagnetic field.
α_{95} = 95 per cent confidence cone about average direction of magnetization [*Fisher*, 1953].
γ_{95} = 95 per cent confidence cone about ancient geomagnetic field direction (see equation 5).

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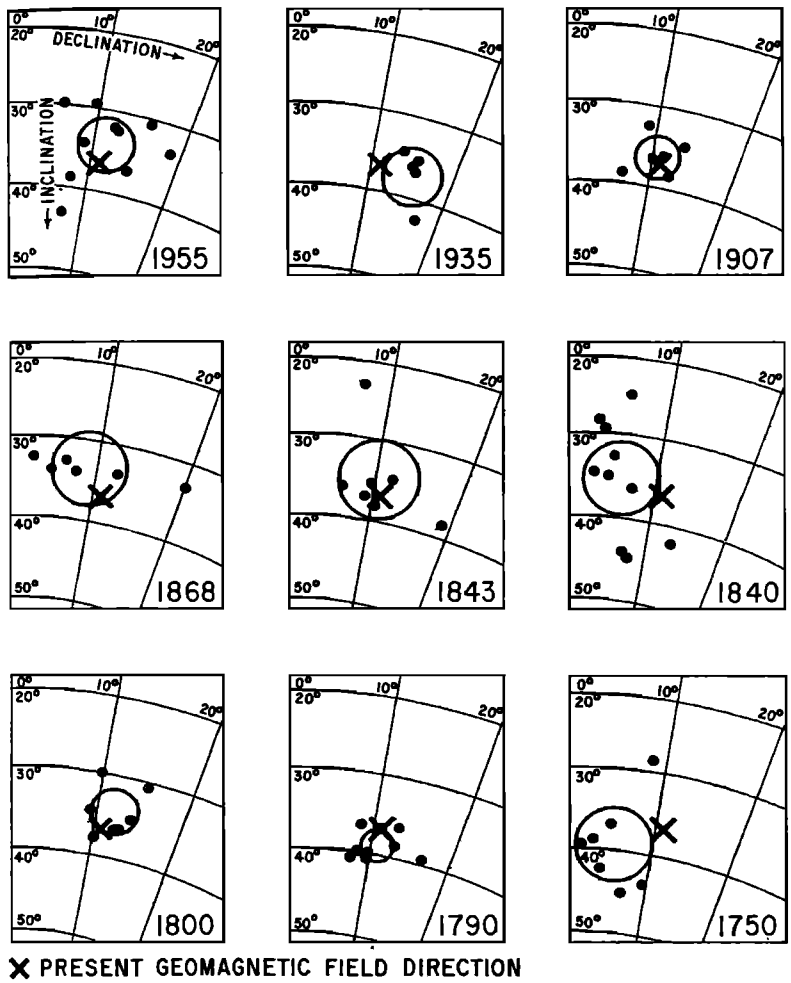


Fig. 4. Directions of remanent magnetization of one specimen of each core after partial demagnetization in 100-oe alternating field. Circles are 95 per cent confidence limits about flow-average directions of magnetization [*Fisher*, 1953].

one specimen from each of the N cores collected from that flow. Figure 3(b) shows similar averages for each flow after all specimens had been partially demagnetized in a 100-oe alternating field. Individual specimen data are shown in Figure 4.

Following the procedures described above, each datum is an average of two demagnetization treatments at 100 oe, the specimen being rotated 180° in the equipment between treatments. The partial demagnetization field of 100 oe was chosen for the following reasons: (1) a substantial part (30 to 80 per cent) of the NRM remains after treatment at this field strength; (2) the IRM experiments show that no IRM is present after treatment in fields this strong; and (3) *Rimbert's* [1959] experiments show that alternating fields of 100 oe are more than sufficient to remove any viscous magnetization acquired in basalts in times as short as 200 years.

The general effect of partial demagnetization on directions of magnetization was small. Only for the 1955 flow was the mean direction of magnetization changed significantly, and this flow was unique in having one core with a widely divergent original direction of magnetization which became nearly parallel to the flow average after partial demagnetization. The initial scatter

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in directions of magnetization of five flows decreased, that of two did not change, and that of the remaining two increased. This variation in response reflects the presence of several components of magnetization undergoing differential changes. Components of magnetization initially parallel to the average direction of magnetization of the flow, including both TRM and possibly some viscous magnetization, are partially demagnetized, increasing the scatter in magnetic directions. Moreover, a new component of magnetization is added owing to imperfections in the demagnetization process, further increasing the scatter. However, partial demagnetization reduces the intensity of softer components of magnetization not initially parallel to the flow average, reducing the scatter. For most of these flows, the latter process appears to predominate.

Two of the three flows extruded since the beginning of direct observatory measurements have average directions of magnetization essentially parallel to the regional geomagnetic field (Table 3). However, the average direction of the 1935 flow differs from that of the regional geomagnetic field by an angle that is small but significant at the 95 per cent probability level. Reasons for this difference are discussed in the next section.

Analysis of Angular Deviations

Rarely is the direction of magnetization of one specimens from a lava flow exactly parallel to the regional geomagnetic field at the time of cooling and, as seen in Figure 4, even flow averages may differ significantly. Thus it is of interest to consider in some detail the sources and magnitudes of angular deviations that arise in a paleomagnetic study before relating these measurements to changes in the regional geomagnetic field direction. The replication measurements and additional observations considered in this section were made to answer, as well as possible, the following questions: (1) How are angular deviations from the regional geomagnetic field direction introduced in a paleomagnetic study of lava flows? (2) How should samples and measurements be distributed to obtain most efficiently a result with a desired accuracy? (3) What natural limitations are there to the ultimate accuracy with which secular variation can be determined from paleomagnetic studies of lava flows?

For convenience, sources of angular deviation may be divided into four main types: (1) deviations added by the experimental procedure, (2) deviations due to changes in the direction of the remanent magnetization after initial cooling, (3) deviations due to a nonparallelism between the original remanent magnetization and the ambient magnetic field, and (4) effects of geomagnetic anomalies of various sizes. The last source is included because most paleomagnetic interest centers on changes in the regional geomagnetic field direction, and the more local anomalies may best be treated as a source of angular deviations. It is useful in this discussion to distinguish five magnetic directions progressing from specimen measurements to regional geomagnetic field direction; the following vector symbols and definitions will be used:

$\mathbf{J}_s$, measured direction of remanent magnetization of a specimen, either before or after partial demagnetization.

$\mathbf{J}_s(t)$, true *in situ* direction of remanent magnetization of a specimen. $\mathbf{J}_s(t_o)$ is the magnetization immediately after initial cooling of the specimen, and $\mathbf{J}_s(t_s)$ is the magnetization direction at the time the specimen was collected.

$\mathbf{F}_s(t)$, direction of ambient magnetic field averaged over the volume occupied by a specimen.

$\mathbf{F}_L(t)$, direction of the magnetic field averaged over the sampled portion of a flow, that is, $\Sigma_N \mathbf{F}_s(t)$, where N is the number of samples collected. $\mathbf{F}_L(t)$ may change between t_o and t_s as defined above.

$\mathbf{F}_R(t)$, regional geomagnetic field direction averaged over a sufficiently large area to exclude local anomalies. The time variation of $\mathbf{F}_R(t)$ is the geomagnetic secular variation in the usual sense.

The magnitude of the angular deviation at various levels will be described by a *Fisher* [1953] precision parameter k and by δ', an approximation to the angular standard deviation that is analogous to a standard deviation of a set of scalars [*Wilson*, 1959].

Level 1: Experimental. $\mathbf{J}_s$ will not be parallel to $\mathbf{J}_s(t_s)$ if errors are made in measuring the remanent magnetization, if angular changes are introduced by the addition of new components of magnetization during the demagnetization

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process, or if errors are made in orienting cores.

Our precision in measuring directions of remanent magnetization is given by $k_P = 15{,}000$, where k is a *Fisher* [1953] precision parameter as defined in equation 2. This number was found by repeated measurements on one specimen over an 8-hour period, using the same magnetometer and removing the specimen from the apparatus between measurements. A value of k_A corresponding to the accuracy of angular measurements is not known exactly; however, an estimate of $k_A = 3600$ was found from repeated measurements of a stable reference magnet by different operators using two magnetometers over a 6-month period.

To evaluate errors due to components of magnetization added by the demagnetization process, we demagnetized one specimen from the 1843 flow and one from the 1935 flow 12 times each in alternating fields of 100 oe. Between successive demagnetizations each sample was rotated 180° in the demagnetization apparatus, and averages were taken of the remanent magnetization directions after successive runs, as before. The precision parameter k_D, calculated for angular variations in these pair averages, after the effects of k_P as previously determined were removed (see equation 3), are 4600 for the specimen from the 1843 flow and 10,300 for the one from the 1935 flow.

If it is assumed that angular errors have a *Fisher* [1953] distribution, an angular standard deviation δ_{rms} can be related to a precision parameter k as follows:

$$\delta_{rms}(\text{radians}) = \left(1/N \sum_{i=1}^{N} \delta_i^2\right)^{1/2}$$

$$\cong (2(N - R)/N)^{1/2} = \delta' \qquad (1)$$

$$k = (N - 1)/(N - R)$$

$$\cong (1 - 1/N)(2/\delta'^2) \qquad (2)$$

where N is the number of unit vectors representing a group of directions, R is the magnitude of the vector sum of the N unit vectors, δ_i is the angle between R and the ith unit vector, and δ' is a convenient approximation to the angular standard deviation as defined in equation 1 [*Wilson*, 1959].

The error in orienting cores is estimated from mechanical considerations to be about 2°. The corresponding precision parameter, k_o, is, from (2), equal to 1640.

If angular variations from different sources are combined, the precision parameter for the total angular variation, k_T, is given by

$$1/k_T = 1/k_1 + 1/k_2 + \cdots \qquad (3)$$

where $k_1, k_2, \cdots$ are the precision parameters for the individual sources of angular variation. Using this equation and the values obtained for k_A, k_D, and k_o, we find the total precision parameter for angular deviations arising at the experimental level to be 900 to 1000.

Level 2: Changes in direction of remanent magnetization after cooling. $\mathbf{J}_s(t_s)$ may not be parallel to $\mathbf{J}_s(t_o)$ either because new components of magnetization have been added or because the original direction of the TRM has changed. New components of chemical, viscous, and lightning-induced magnetization, which are common in older rocks, are absent in the historic Hawaiian flows after partial demagnetization to 100 oe. Changes in the original magnetization may be caused by changing stresses, especially if the rocks are anisotropic, but such effects are probably negligible in these unburied extrusives. In the historic Hawaiian lava flows the largest source of angular variation at this level probably arises from rotation of blocks of lava after cooling. Because of the short geological history of these flows, small random rotations due to jostling of the cooler parts of the flow during extrusion are probably more important than faulting or tilting of the underlying rock. No unique evaluation of the amount of variation introduced by these small rotations is possible; however, the effects will appear as part of the core-to-core variations which are evaluated below.

Level 3: Nonparallelism of original magnetization and ambient magnetic field directions. $\mathbf{J}_s(t_o)$ will not be acquired in a direction exactly parallel to $\mathbf{F}_s(t_o)$ if ferromagnetic minerals are distributed in layers or if they have a preferred crystallographic orientation; however, no pronounced anisotropy was observed in thin sections prepared from these flows. Thermal gradients developed during cooling have also been suggested by *Kuzhelov and Krutikhovskaya* [1960] as a possible source of angular error at this level. This effect appears to be absent from these lavas, since the theory of Kuzhelov and

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Krutikhovskaya predicts that in bodies with horizontal isothermal cooling surfaces the inclination of the TRM will be less than the inclination of the ambient magnetic field, a condition which is not observed in the historic Hawaiian flows.

Another source of angular deviation at this level arises from the statistical expectation of the magnetization of a finite number of randomly oriented ferromagnetic domains that were not magnetized parallel to the ambient magnetic field during cooling [*Irving et al.*, 1961]. The saturation remanent magnetization of the historic Hawaiian lavas is typically 50 times larger than the natural TRM (Table 1), indicating that on cooling most ferromagnetic domains assume a random rather than a preferred orientation. On the basis of a simplified statistical model, *Irving et al.* [1961] estimated that the intensity of the component of magnetization due to this cause may be as large as 10 per cent of the TRM acquired when the lava cooled. They also interpreted certain demagnetization experiments as indicating that the intensity of the random component is from 0.6 to 36 per cent of the intensity of the natural remanent magnetization in the lavas they studied.

Vectorial addition of a random component with intensity J_R to an ideal stable magnetization of intensity J_S which is parallel from specimen to specimen throughout a flow would introduce a specimen-to-specimen angular standard deviation δ'_R, given by

$$\delta'_R(\text{radians}) \cong 0.8(J_R/J_S) \qquad (4)$$

the approximation being valid for small values of J_R/J_S. The value of $J_R/J_S = 0.1$ suggested by *Irving et al.* [1961] would result in an angular standard variation between specimens of 4½°, corresponding to a precision parameter, k_R, of 310.

Since random components this large could account for most of the angular variation observed in the historic Hawaiian lavas, the following analysis was made to estimate a maximum value for the random component in these flows. The average angular variation between all specimens from all cores collected from the 1955 and 1790 flows was calculated using the two-level statistical method of *Watson and Irving* [1957]. To eliminate preferred alignments of domains having low coercive force, each specimen was first demagnetized at 100 oe. In contrast with other demagnetization experiments, each specimen was demagnetized only once to avoid the loss of some of the random component through averaging. The within-core or specimen-to-specimen precision parameter for the 1955 flow was 1330, and for the 1790 flow it was 4260; corresponding values of the angular standard deviation are 2.2° and 1.2°. These values include effects of measurement precision, components added by the demagnetization process, and natural specimen-to-specimen variation due to other causes. It is therefore probable that the ratio of J_R/J_S for the historic Hawaiian lava flows is considerably less than the maximum value of 2½ to 5 per cent found from these experiments. Thus the random component accounts for only a small part of the observed intraflow variation.

Level 4: Intraflow magnetic field anomalies. Variations in directions of magnetization between different cores are in part due to variations in the direction of the magnetic field from place to place within a cooling lava flow. Such intraflow variations of the magnetic field are expected on Hawaii because flows are usually extruded onto an irregular terrain composed of rather strongly magnetized lavas.

An upper limit may be estimated for angular deviations due to such intraflow magnetic anomalies from the following observations. When the sample cores were oriented, the departure ΔD of the declination of the local magnetic field from the declination of the regional field was measured at a point about 10 cm from the surface where the core was drilled. A single ΔD value of 8° was recorded, but most were less than 3°. Standard deviations of the ΔD values calculated for all the cores from individual flows vary from 0.9° to 4.2° (Table 3), with an average value of 2.4°. If it is assumed that inclinations (which were not measured) vary a similar amount from core to core, the average precision parameter for the intraflow field variations will be 1700. More complete magnetic field surveys over volcanic terrain in Hawaii at heights about equal to half the average thickness of lava flows are needed for a more precise direct evaluation of this quantity.

Variations in directions of magnetization from core to core in a lava flow may also set an upper limit to variations at this level. In order to separate core-to-core from specimen-to-specimen

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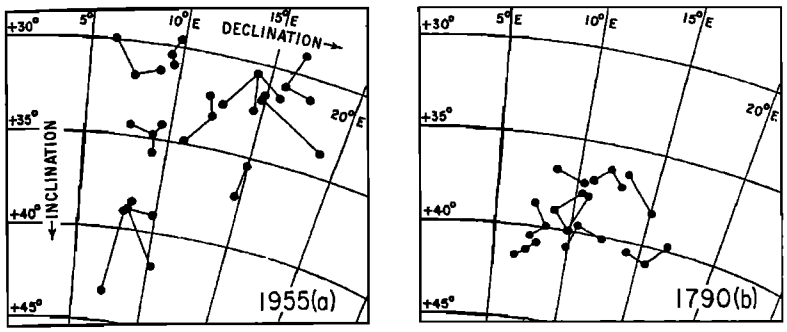


Fig. 5. Directions of remanent magnetization of all specimens from the 1955 (*a*) and 1790 (*b*) flows after partial demagnetization to 100 oe. Lines join specimens from the same core.

variations, two-level analyses [*Watson and Irving*, 1957] were made of all specimens from the 1955 and 1790 flows. To minimize the effects of small components added during demagnetization, averages after double demagnetization at 100 oe were used. Results are shown in Figure 5.

The core-to-core variations of 4.5° and 2.6° shown in Table 4, which summarizes the two-level analysis, establish only an upper limit to intraflow variations in magnetic field directions, since these figures also include the effects of orientation errors (level 1), errors from small-scale rotations (level 2), and possibly some effects due to anisotropy (level 3). The specimen-to-specimen variations in Table 4 include effects from all other causes previously described and may also include some effects due to anisotropy.

Level 5: Local and regional fields not parallel. Since large-scale magnetic anomalies in volcanic regions commonly have dimensions of the order of tens of kilometers, an entire lava flow may cool in a region where $\mathbf{F}_L(t_o)$ is not parallel to the regional geomagnetic field, $\mathbf{F}_R(t_o)$. Magnetic anomaly maps needed for evaluating such angular variations on Hawaii are not available. However, the detailed magnetic anomaly data for the seven Izu volcanic islands extending south from Tokyo Bay [*Rikitake et al.*, 1951; *Yokoyama*, 1957] may be used in estimating such variations on Hawaii.

TABLE 4. Two-Level Analysis of Angular Deviations between Specimens from the Same Core and between Cores from the Same Flow

Date of Flow	Number of Cores	Number of Specimens	Specimen-to-Specimen Variation		Core-to-Core Variation	
			k_S	δ'_S	k_C	δ'_C
1955	10	31	2460	1.6°	320	4.5°
1790	9	24	4800	1.2°	1000	2.6°

One of the larger islands, Ooshima, has a major magnetic anomaly extending over the entire island which approximates in form the anomaly field of a buried dipole. Near the central summit of the island, inclination anomalies as large as 3½° persist over regions several kilometers long. Declination anomalies are of smaller amplitude but of greater lateral extent. The magnetic anomaly maps of Ooshima prepared by *Rikitake et al.* [1951] and *Yokoyama* [1957] were used in determining the standard deviation of angular departures between local and regional geomagnetic field directions for 96 points evenly distributed over the island at 1-km centers. The resulting value of the standard deviation ($\delta'_L = 2.0°$) corresponds to a precision parameter k_L of 1640.

These results cannot be extended to Hawaii without first considering the effects of differences in the geomagnetic field strength at the two islands and the differences in average intensities of magnetization. The average intensity of the 152 lava flows sampled on Hawaii is 0.6 $\times 10^{-3}$ emu/cm^3; that on Ooshima is 1.0 $\times 10^{-3}$ emu/cm^3, as estimated both from the magnetic anomalies [*Yokoyama*, 1957] and from direct measurements coupled with stratigraphic considerations [*Yukutake*, 1961; *Nakamura*, 1960]. Taking into account the 20 per cent lower geomagnetic field intensity in Hawaii and the difference in average remanent magnetization, we ob-

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tain a value of 1° to 1½° for δ'_L, the standard deviation between $\mathbf{F}_L(t)$ and $\mathbf{F}_R(t)$ on Hawaii. A more precise evaluation of this parameter must await detailed geomagnetic field surveys on Hawaii.

Because the spatial distribution of magnetic anomalies on Hawaii in the past is unknown, their effect can only be considered statistically as an additional error which should be used to broaden the confidence interval about the average direction of magnetization of a lava flow in order that the confidence interval will correspond to the accuracy with which $\mathbf{F}_R(t)$ rather than $\mathbf{F}_L(t)$ is known. To a good approximation, if $\alpha_{(1-P)}$ is the confidence limit for the average direction of magnetization of a flow at probability level P [*Fisher*, 1953], and if k_L is the precision parameter corresponding to magnetic anomalies, the statistical confidence limit for the ancient regional field direction, $\gamma_{(1-P)}$, is given by

$$\cos \gamma_{(1-P)} \cong \cos \alpha_{(1-P)} + (1/k_L) \log_e P \quad (5)$$

Values of γ_{95} for the historic Hawaiian flows are listed in Table 3, using $k_L = 4400$ ($\delta'_L = 1.2°$). It is of interest to note that the mean direction of the 1935 flow differs significantly from the known direction of the regional geomagnetic field at 95 per cent probability if α_{95} is used as the criterion, but it does not differ if γ_{95} is used.

Discussion. The replication experiments and other observations discussed in this section and summarized in Table 5 lead to the following conclusions: (1) Increasing the accuracy of individual measurements would not substantially increase the over-all accuracy; for example, decreasing δ'_A by 50 per cent would achieve only a 2 per cent reduction in γ_{95}. (2) Very little increase in over-all accuracy would be achieved by reducing the scatter in directions of magnetization introduced by the demagnetization process; for example, a 50 per cent decrease in the standard deviation of the latter would cause only a 2 per cent reduction in γ_{95}. (3) The random component of magnetization due to statisti-

TABLE 5. Summary of Replication Experiments

Description of Error	Observations Made	Standard Deviation		Precision Parameter	
		Quantity	Value	Quantity	Value
Measurement precision (level 1)	Replicate measurement of one specimen over 8-hour period	δ'_P	0.7°	k_P	15,000
Measurement accuracy (level 1)	Replicate measurement of reference magnet over 6-month period	δ'_A	1.3°	k_A	3,600
Components added by demagnetization in 100-oersted field (level 1)	Replicate double demagnetization of one specimen in 100-oe field	δ'_D	1.2° to 0.8°	k_D	4,600 to 10,300
Orientation errors (level 1)	Estimated from mechanical considerations	δ'_O	2.0°	k_O	1,640
Random component due to statistical expectation of net moment of finite number of randomly oriented domains (level 3)	Variations between specimens from the same core after one demagnetization at 100 oe (no averaging)	δ'_R	<2.2° to 1.2°	k_R	>1,330 to 4,260
$\mathbf{F}_S$ not parallel to $\mathbf{F}_L$; intraflow variations in direction of ambient field (level 4)	Intraflow changes in declination where cores were collected	δ'_F	<2.4°	k_F	>1,700
Core-to-core variations: block rotations (level 2), intraflow ambient field variations (level 4), and orientation errors (level 1)	Two-level analysis of all specimens from several flows after double demagnetizations at 100 oe	δ'_C	4.5° to 2.6°	k_C	320 to 1,000
$\mathbf{F}_L$ not parallel to $\mathbf{F}_R$: local magnetic anomalies extending over an entire lava flow (level 5)	Comparison with geomagnetic field surveys on Ooshima Island	δ'_L	1.2°	k_L	4,400

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cal randomization effects has an intensity no greater than 5 per cent of the TRM and very probably has considerably less; its contribution to angular errors is minor. (4) Most intraflow angular variations enter at the core-to-core level and are due to errors in orienting cores, to rotations in blocks of lava after cooling, to intraflow differences in the direction of the ambient magnetic field at the time the flow cooled, and possibly to other unknown causes. The values of δ'_θ, the estimated value for orientation errors, and δ'_F, as estimated from the ΔD values, suggest that a substantial part of the core-to-core variation arises from these two causes; a useful increase in over-all accuracy may be made by improving the accuracy of orienting cores. (5) Double demagnetization treatment of these high-coercive-force specimens does not increase over-all accuracy enough to warrant its use except as an occasional check on the demagnetization efficiency. (6) Since the core-to-core angular variations are much larger than those between specimens from the same core (Figure 5), little increase in over-all accuracy results from measuring several specimens from the same core (cf. intraflow k_C values for the 1955 and 1790 flows in Tables 3 and 4). (7) The ultimate accuracy that can be achieved in a paleomagnetic study of secular variation in a volcanic region is limited by the magnetic anomalies existing over the old terrain where the lava flows cooled. On Hawaii the smallest possible confidence limit at the 95 per cent probability level is estimated to be 2½°. Theoretically this error might be reduced by calculating the ancient magnetic anomaly over the old volcanic terrain as reconstructed from geological evidence, but in general this would be very difficult. (8) The optimum number of samples to be collected in a paleomagnetic study of lava flows depends on the accuracy required, the value of k_L for the region, and the amount of intraflow angular variation. The dependence of γ_{95} on N, the number of samples, and on k_C, the intraflow precision parameter, for a value of $k_L = 4400$, is shown in Figure 6. The optimum number of samples to be collected from lava flows with intraflow precision parameters in the range 200 to 800 appears to be about six, a number which ensures that the γ_{95} values will be from 3° to 4½°. Little additional accuracy is gained by collecting inordinately large numbers of samples; if the

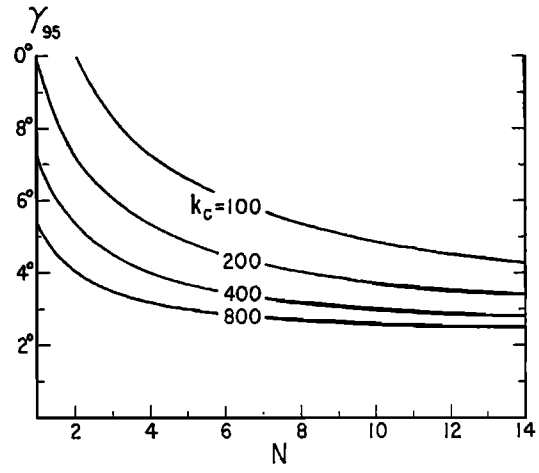


Fig. 6. Curves showing the number of samples, N, required to obtain a given 95 per cent confidence limit, γ_{95}, for different intraflow precision parameters, k_C.

number of samples is increased from six to twelve in the above example, the range of γ_{95} values is 2½° to 3½°. Optimum accuracy is undoubtedly achieved from a given number of samples if they are spaced over a lateral distance of several hundred meters; however, results for the 1935 and 1907 flows suggest that in most cases little accuracy is sacrificed if the spacing is somewhat smaller.

Secular Variation on Hawaii

Values of declination and inclination determined paleomagnetically from the nine historic Hawaiian lava flows are plotted in Figure 7, along with declination and inclination curves recorded at Honolulu during the past 50 years, and extrapolated to the sampling area using the 1945 regional field gradients [*Vestine et al.*, 1948]. Curves of declination and inclination recorded at London, England, where there has been a rather typical rate of secular variation, and at Ascension Island, where the secular variation is unusually large, are also shown for comparison.

Secular variation curves determined from observatory records that extend back several hundred years are usually quite smooth, with average rates of change not exceeding 2° per decade and with periods of at least 200 years between major inflection points. If secular varia-

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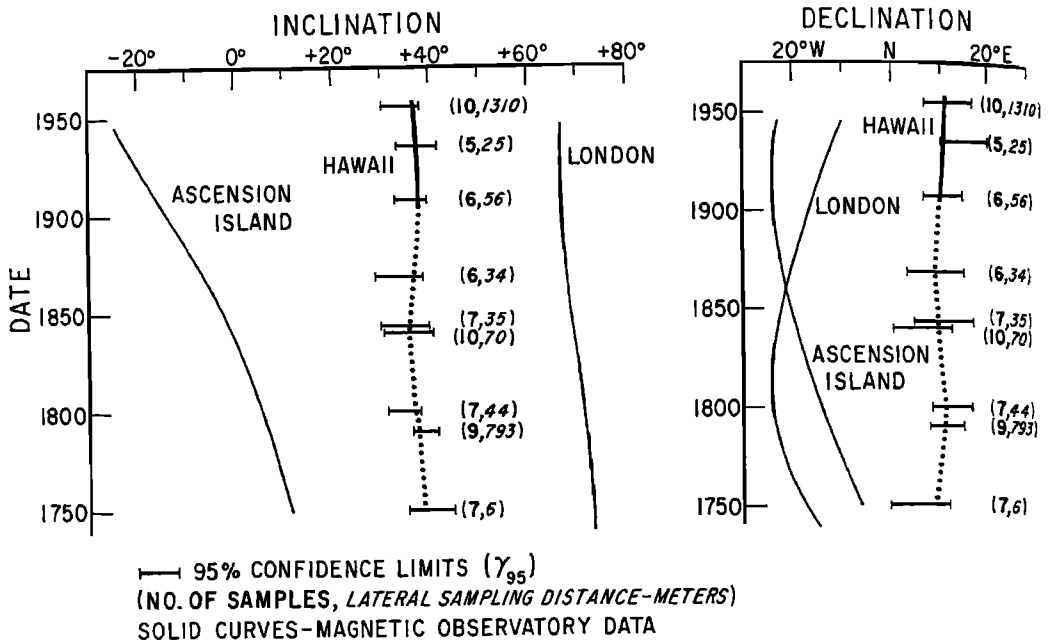


Fig. 7. Secular variation on Hawaii since 1750.

tion on Hawaii has behaved similarly, the values of declination and inclination since 1750 should lie close to the dotted curve shown in Figure 7.

The anomalously low rate of secular variation in the Pacific Ocean area during this century has long been noted [*Fisk*, 1931], and these paleomagnetic results from the historic Hawaiian lavas imply that this anomalously low rate has persisted since 1750. This is also consistent with paleomagnetic results from prehistoric Hawaiian lavas [*Doell and Cox*, 1961] which indicate that rates of secular variation on Hawaii may have been anomalously low for extended periods of time during the Pleistocene.

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